(a)

1980s – Early applications of artificial neural networks (ANNs) in hydrological and waterresource modeling

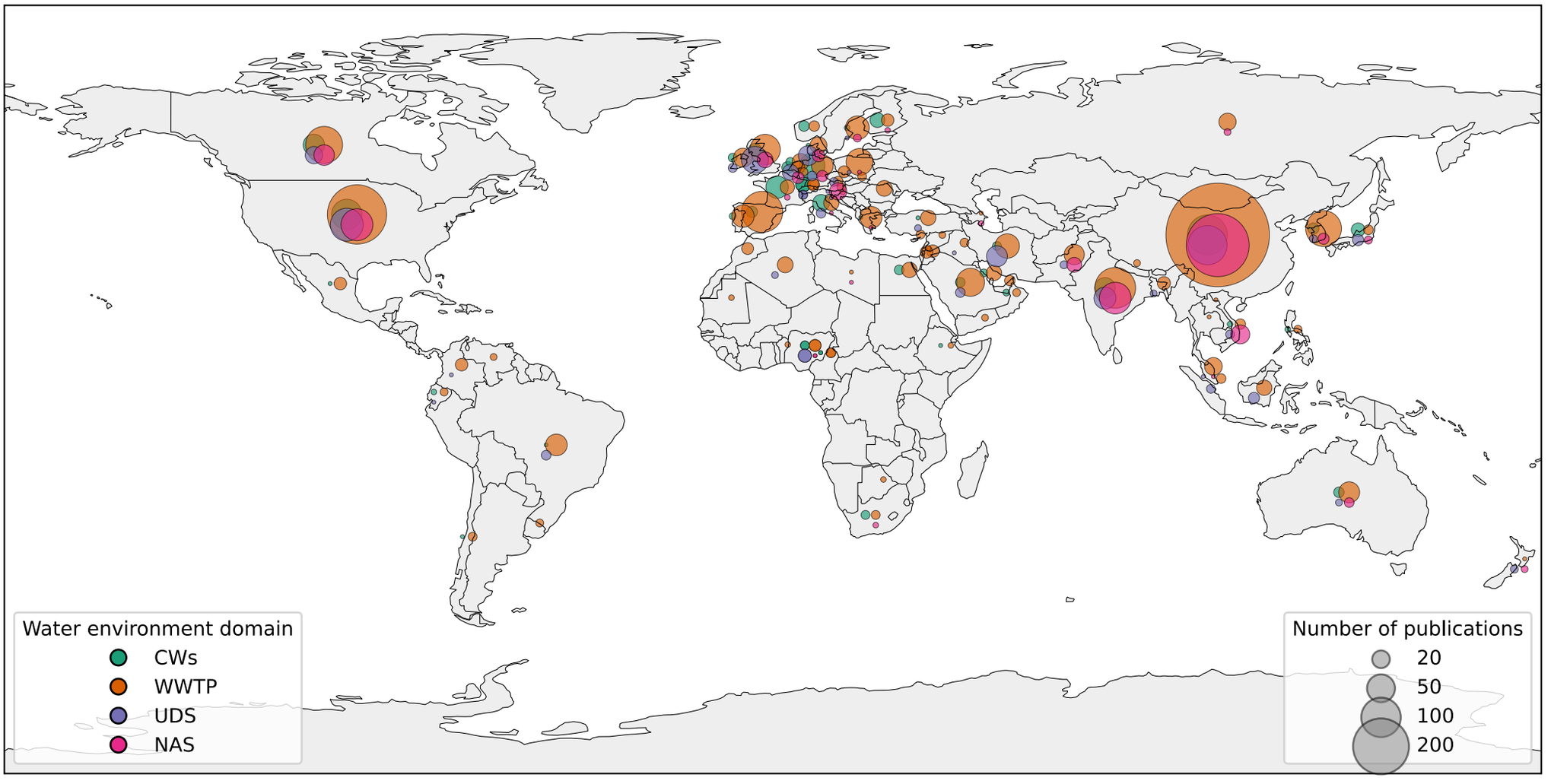
2005 – Neural network–based groundwater level forecasting and subsurface hydrological modeling

2020s – Rapid development of AI-driven water environmental modeling and prediction

1990 2000 2010 2020

1990s – Expansion of AI techniques (ANNs and genetic algorithms) for water quantity and quality prediction

2017–2020 – Emergence of deep learning in water resources and hydrological sciences

Recent years – Accelerated growth of AI applications across water environments and tasks

(b) Global distribution of AI applications across major water environments (1989–2025)